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RESEARCH ARTICLE

Efficient Unified Framework for Difficulty-Controllable Question Generation: Large Language Models Enabling Active Learning Labeling and Prompt-Based Low-Rank Adaptation

XIAOHUI DONG AND ZHENGLUO LI
College of Computer Science and Engineering, Northwest Normal University
Corresponding author: Xiaohui Dong (dxh@nwnu.edu.cn)

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ABSTRACT The task of Difficulty-Controllable **Q**uestion **G**eneration (**QG**) in reading comprehension is of great significance for improving the educational applicability of automatically generated questions. However, existing research faces two core challenges: 1) lack of unified difficulty quantification standards and annotated datasets leads to limitations in the quality and applicability of generated questions; 2) Difficulty-Controllable generation relies on a single feature or high-cost simulation systems, making it difficult to balance diversity and practicality. To address these issues, we propose **DL-CQG**, a unified framework for **D**ifficulty **L**abeling and **C**ontrollable **QG**. This framework defines difficulty levels based on Bloom's theory, employs **L**arge **L**anguage **M**odel-**A**ugmented **A**ctive **L**earning (**LLM-AAL**) to achieve efficient difficulty annotation, and fine-tuning LLMs with prompts and LoRa to adapt to the Difficulty-Controllable QG task. Experiments show that on the EQG-RACE dataset, the LLM-AAL framework can train a difficulty evaluation model with an accurate rate of 0.85, using only a small number of samples, which significantly reduces annotation costs. In the QG task, the adapted Llama-3.2-3B-Instruct model outperformed all baseline methods across metrics, achieving scores of 45.20 (BLEU-4), 50.01 (ROUGE-L), and 81.26 (BERT Score), with an 0.824 accuracy rate in question difficulty control. Our proposed framework establishes a unified paradigm for Difficulty-Controllable QG under low-resource conditions.

INDEX TERMS Natural language generation, question generation, large language model, active learning, prompt-tuning.

I. INTRODUCTION

Question **G**eneration (**QG**) for reading comprehension is a challenging task in the field of natural language processing, aiming to automatically generate questions related to a given reading passage. This technology exhibits broad application prospects in intelligent education scenarios, such as intelligent tutoring systems [1], writing support systems [2], and knowledge assessment systems [3], [4], [5]. Due to the rapid advancement of natural language processing and deep learning technologies, remarkable progress has been achieved in research on automatic question generation [6], [7].

Difficulty-Controllable QG is a relatively new task [8]. The goal is to automatically generate questions based on the specified difficulty level to meet the needs of different learners. Therefore, this task is of great significance for

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improving the educational applicability of automatically generated questions, but relevant research remains extremely limited [9]. Reference [10] first formalized Difficulty-Controllable QG, using two machine reading comprehension models to classify questions as 'hard' or 'easy' based on their accuracy, then controlling generation difficulty through hidden states. However, this method is empirically driven and lacks theoretical basis. Reference [3] adopted the same idea, simulated subjects with 60 question-answering systems, and quantified question difficulty based on **I**tem **R**esponse **T**heory (**IRT**). On this basis, the pre-trained model T5 was fine-tuned to generate question-answer pairs that meet the difficulty target. Reference [4] estimated the difficulty based on the popularity of named entities in the knowledge graph, while references [11] defined question difficulty as the number of reasoning steps required to answer a question and gradually rewrite simple questions as complexity questions through question rewriting techniques. There is also research [12] that regards questions requiring multi-step reasoning to answer as deep questions and uses sentence-level rewards of reinforcement learning to enhance the depth of generated questions.

It can be seen that Difficulty-Controllable QG faces two main challenges: the first is how to quantify question difficulty and use it as the basis for dataset annotation, then training reliable neural network models. The second is the design of the difficulty control mechanism for QG. Due to the lack of a unified standard for measuring difficulty and annotated datasets, the first challenge is the main bottleneck in current research. Existing methods have adopted two approaches to address this issue. One is to simulate the subjects by question-answering system and measure the difficulty of the questions by the correct rate of the subjects' answers. Although this approach can leverage IRT to obtain the true difficulty of questions, it is expensive to construct a considerable number of questions answering systems. For example, a simple binary classification task requires at least 100-150 question answering systems to simulate the subjects in order to obtain reliable results using IRT. Therefore, this method has great limitations. The other approach is to simplify question difficulty to a single feature, such as the number of reasoning steps or the type of interrogative words, which may lead to a lack of diversity in the generated questions. Regarding the second challenge, embedding difficulty labels into the neural network decoder to manipulate the QG process [10], or using reinforcement learning to optimize the target generation [12], and fine-tuning **L** arge **L**anguage **M**odels (**LLM**s) are the main methods used in existing research. Relatively speaking, since LLMs possess powerful language understanding and generation capabilities, the work of fine-tuning LLMs to adapt to different QG tasks has gradually become the mainstream approach [13], [14], [15].

To alleviate the above limitations, we propose a unified framework for Difficulty-Controllable QG, named **D**ifficulty **L**abeling and **C**ontrollable **QG** (**DL-CQG**). This framework defines question difficulty levels based on Bloom's theory and consists of two parts: question difficulty evaluation and Difficulty-Controllable QG. We first employed the **LLM-A**ugmented **A**ctive **L**earning (**LLM-AAL**) framework for the iterative fine-tuning of Qwen2.5-3B-Instruct, building an automated question difficulty evaluator for annotating the EQG-RACE dataset. To optimize sample quality, we implement maximum entropy sampling and difficulty-balanced filtering are implemented, ensuring the representativeness and uniform distribution of annotations. Subsequently, we fine-tune LLMs (such as Llama-3.2-3B-Instruct) with prompts and LoRa to adapt the Difficulty-Controllable QG task.

We evaluated DL-CQG on the EQG-RACE dataset. First, during the LLM-AAL iterative process, we initially selected 50 samples for annotation, and the sample size doubled in each subsequent iteration. When the sample size reached 400, 600, and 800 using maximum entropy sampling, the accuracy of the **Diff**iculty **E**valuation (**Diff-E**) model exceeded 0.85 in all cases, indicating that the Diff-E model can replace manual annotation. Second, the QG experiments showed that the adapted Llama-3.2-3B-Instruct outperformed other baseline models in all metrics, with a difficulty accuracy rate of the generated questions reaching 0.824. Ablation experiments and human evaluations further verified the effectiveness of our proposed method.

The main contributions of this paper are summarized as follows:

(1) We propose a unified framework DL-CQG for Difficulty-Controllable QG, which leverages Bloom's theory to define question difficulty levels, enabling both difficulty annotation and controllable generation through fine-tuning LLM with prompts and LoRa.

(2) We trained a Diff-E model that can replace manual question difficulty evaluation using the LLM-AAL framework. This method can train a question Diff-E model with a small number of samples, providing a solution to the lack of difficulty-annotated datasets.

(3) The experimental results on the EQG-RACE dataset demonstrate that our method significantly outperforms baseline methods when evaluating the quality of generated questions, achieving 0.824 accuracy in difficulty prediction.

The structure of the remainder of this paper is as follows: Section II reviews related work on QG and Difficulty-Controllable QG. Section III elaborates on the proposed framework and the design principles of its various modules. Section IV details the experimental setup, including difficulty annotating datasets, evaluation metrics, baseline methods, and implementation specifics. Section V presents the experimental results and provides a comprehensive analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORK

A. QUESTION GENERATION

The study of QG can be traced back to the 1970s [16]. Early QG methods were primarily rule-based

or template-driven, converting declarative text into interrogative sentences through manually designed rules or templates [3], [11]. These methods required significant manual intervention, and the diversity and quality of the generated questions heavily depended on the coverage of predefined rules or templates. Typically, they could only generate factual questions, exhibiting obvious limitations [16].

With the rise of deep learning, QG research gradually shifted toward data-driven methods. Reference [17] were the first to introduce the encoder-decoder (Seq2Seq) architecture to the QG task, employing attention mechanisms to enhance generation performance. Subsequently, models based on recurrent neural networks (RNN), long short-term memory networks (LSTM), gated recurrent units (GRU), and Transformers became mainstream in QG research [18], [19], [20] proposed an answer-aware QG method, using a copy mechanism to address rare-word problems while explicitly modeling the relationship between answers and context. To overcome the limitations of single-sentence question generation, [21] proposed a paragraph-level QG model, employing gated self-attention mechanisms to copy key information from the entire document. Additionally, researchers explored strategies such as question-type prediction [18], answer-position features [22], self-attention mechanisms [23], and weakly supervised enhancement [24] to further improve the quality of generated questions.

In recent years, advanced methods such as pre-trained language models (PLMs) [25], [26], variational autoencoders [24], graph-based frameworks [27], and generative adversarial networks [28] have been widely applied to QG tasks. These methods can more effectively capture semantic information and construct contextual representations among input text, answers, and questions, thereby significantly improving the quality and diversity of generated questions [26], the literature [29] and [30] focuses on optimizing question word generation by predicting question word types during the decoding phase to improve the accuracy of generated questions. Meanwhile, the introduction of LLMs has brought new insights to this field. For example, Ma et al. [31] leveraged LLM-enhanced decoupled learning to significantly improve the performance of personalized recommendation systems. This progress also indirectly confirms the broad potential of adopting LLM-based interaction-enhanced learning in improving the quality of personalized courses.

B. DIFFICULTY-CONTROLLABLE QUESTION GENERATION

In real-world application scenarios, people's demands for problem difficulty are diverse. With the rapid development of information retrieval technologies, intelligent dialogue systems, and educational technologies, how to automatically generate questions with clear objectives and adaptable difficulty has become a key research direction. This technological breakthrough can not only significantly optimize user experience but also provide crucial support for core application scenarios such as online education platforms, knowledge graph construction, and multimodal human-computer interaction. Taking the education field as an example, existing research [32] has shown that learners at different cognitive levels require questions designed with varying difficulty; moreover, providing questions of appropriate difficulty has been proven to effectively enhance learners' performance [33]. Recent work indicates that question-answering models have made significant progress in addressing simple questions [34]. These models can improve their performance by leveraging diverse training datasets derived from QG methods [35]. Therefore, generating Difficulty-Controllable questions is key to further expanding the applicability of QG in practical domains.

Generating questions that conform to specified difficulty labels faces multiple challenges. First, existing difficulty label embeddings only treat them as vector inputs to the encoder [10], [36]. However, such embeddings contain limited information and struggle to converge when faced with diverse inputs, leading to a tendency for the decoder to ignore the vector—a phenomenon known as ''attribute collapse'' [37], [38]. Additionally, due to the lack of parallel corpora, traditional methods cannot learn the causal relationship between difficulty labels and generated questions, resulting in poor difficulty control—failing to generate diverse questions or produce differentiated outputs for the same subgraph under the same difficulty label. [6] adopted a knowledge-graph-based Difficulty-Controllable QG model using soft templates and counterfactual reasoning to strengthen the causal association between difficulty labels and generated questions. However, the types of difficulty labels are scarce, and the difficulty annotation method lacks interpretability. This paper proposes a unified framework for difficulty annotation and controllable QG, aiming to define question difficulty levels based on Bloom's theory, employ LLM-Augmented Active Learning for efficient difficulty annotation, and generate questions of specified difficulty through prompting and LoRa fine-tuning of LLMs.

III. FRAMEWORK DESCRIPTION

Our method defines question difficulty levels based on Bloom's theory [39]. The core of Bloom's theory is the process from lower-order thinking to higher-order thinking. Each question type in Bloom's taxonomy can be further divided into three common difficulty levels: easy, medium, and hard [40]. Specifically, easy corresponds to recognition and recall in Bloom's theory, medium corresponds to understanding and hard corresponds to the other categories. Our unified framework for Difficulty-Controllable QG, DL-CQG, consists of two main steps. In the first step, we train a Diff-E model that can replace human evaluation of question difficulty using the iterative process of the LLM-AAL framework and use this model to annotate question difficulty levels. In the second step, we input the context, answer, difficulty level, and the prompt words designed into the LLM,

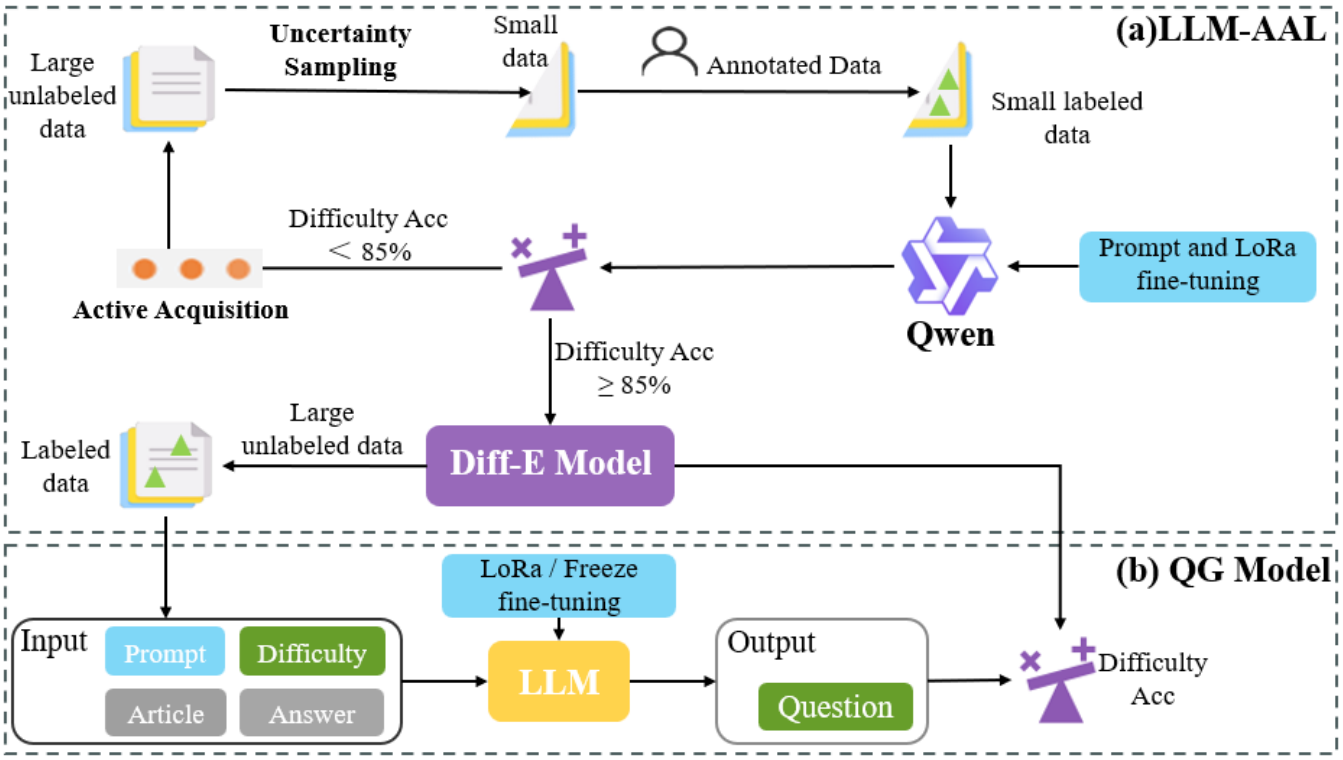


FIGURE 1. The LLM-AAL framework (a) and the difficulty-controllable QG framework (b). Here, Acc denotes accuracy.

and fine-tune the LLM with LoRa to generate questions that match the specified difficulty level. The overall architecture of our method is shown in Figure 1.

A. LLM-AAL FRAMEWORK

To reduce annotation costs and conserve computational resources, we adopted the LLM-Augmented Active Learning collaborative loop to train the Diff-E model, as proposed in reference [41]. The base model for Diff-E is Qwen2.5-3B-Instruct. The iterative process of LLM-AAL is shown in Figure 1(a). This method comprises the following three core steps.

Step 1. Data processing: To enhance the efficiency and effectiveness of the active learning process, we first select annotation samples using uncertainty sampling [42] and then filter samples with different difficulty labels to ensure that the annotation samples are representative and evenly distributed.

Uncertainty Sampling: We use the Qwen2.5-3B-Instruct model to make predictions and calculate token-level uncertainty using entropy. The data samples are sorted in descending order of entropy values, and the top 0.02 of samples with the highest uncertainty are selected as candidate samples for annotation. The formula for calculating entropy is:

$$H(i) = -\sum_{c=1}^{C} p(c|i) \log p(c|i) \tag{1}$$

For each token i, $p(c|i)$ represents the probability of category c, and C is the total number of difficulty level categories (C=3).

Next, we calculate the average entropy $H_{avg}(sample)$ for each sample, using the following formula:

$$H_{avg}(sample) = \frac{1}{|sample|} \sum_{i \in sample} H(i) \tag{2}$$

Subsequently, by manually selecting samples of varying difficulties to ensure an even distribution across each difficulty level, we invited three educational assessment experts to annotate the samples with difficulty levels based on Bloom's theory. To ensure consistency in annotation, we employed Fleiss' Kappa coefficient (which is suitable for multiple raters evaluating the same set of discrete classification tasks) to measure inter-rater reliability, ensuring that the Kappa value exceeded 0.7 (Kappa $>$ 0.7, indicating substantial agreement among raters). For any inconsistent annotations, the raters reached consensus through discussion and updated the labels accordingly.

Step 2. Fine-tuning LLM: We use the annotated samples from Step 1 as the input to train the Diff-E model, with the model output being the difficulty label of the question (easy/medium/hard). We fine-tune the Qwen2.5-3B-Instruct model using a prompt and LoRa.

Prompt: As shown in Figure 2, we employ prompts to guide the model in generating difficulty labels that align with the target. By setting the model's role, clearly defining the difficulty level criteria, specifying the task requirements, and conducting classification validation (gap detection, scope verification, difficulty level confirmation), we ensure that

the model outputs difficulty labels that meet the research objectives.

LoRa fine-tuning: Utilizing low-rank decomposition techniques, we introduce trainable low-rank matrices while freezing the original parameters of the pre-trained model, achieving parameter-efficient adaptation. This method significantly reduces the computational complexity and storage requirements during the training process while maintaining model performance, thereby enhancing the training efficiency of the model.

Step 3. Diff-E Model Construction Process: In this step, we use the Accuracy metric to measure the consistency between the model's predicted labels and the true labels. The formula for calculating Accuracy is:

$$Accuracy = \frac{N_{correct}}{N_{total}} \tag{3}$$

Here, $N_{correct}$ denotes the number of correct predictions, and N_{total} represents the total number of samples.

If the model's Accuracy is below 0.85, we iterate Steps 1 and 2. When the model's Accuracy reaches or exceeds 0.85, and there is no significant improvement in performance after training with additional samples, we consider the fine-tuned model capable of replacing manual annotation for question difficulty labeling. In other words, we have obtained the Diff-E model for questions, which can be used to annotate the difficulty of the dataset.

Component	Prompt
Document	{Article},{Question},{Answer}
Role	You are an expert in educational evaluation.
Difficulty level description of the questions	Easy-Level Questions: Recognition、Recall Medium-Level Questions: Understanding Hard-Level Questions: Application、Analysis、Creation、Evaluation
Task Requirements	Based on the provided Article, Question, and Answer, analyze the cognitive level required for the question and evaluate its difficulty according to Bloom's Taxonomy. The output should be one of the following: easy, medium, or hard.
Classification verification	Gap detection、Scope verification、Difficulty level confirmation

FIGURE 2. The prompt adopted in LLM-AAL.

B. QG MODEL

We employ prompts combined with fine-tuning of LLMs (BLOOM, Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct) to achieve Difficulty-Controllable QG. The prompt design includes role setting, task description, and task requirements, with the detailed design shown in Figure 3. We adopt the following two fine-tuning methods:

(1) LoRa fine-tuning: As described in Step 2 of Section III-A.

(2) Freeze fine-tuning: By freezing some or all of the original parameters of the pre-trained language model, only a small number of additional trainable modules (such as adaptation layers) or specific network layers are updated to achieve efficient model adaptation. This fine-tuning method significantly reduces computational overhead during training, enabling fast and efficient model fine-tuning in resource-constrained scenarios.

Our Difficulty-Controllable QG model framework is shown in Figure 1(b). We input the article, answer, difficulty label, and prompt into the DL-CQG framework, and the output is a question with the specified difficulty. Drawing on reference [43], given a difficulty label d_i, prompt pr', article c, and answer a, the QG task can be defined as generating a question $\hat{y}$:

$$\begin{aligned}\hat{y} &= \arg\max_{y} P(y \mid c, a, d_i, pr') \\ &= \arg\max_{y} P(y \mid x)\end{aligned} \tag{4}$$

Here, $P(y|x)$ represents the conditional log-likelihood of predicting the sequence y given the input $x = (c, a, d_i, pr')$.

The training process is as follows: Given a training corpus $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{|D|}$, where $x = (c, a, d_i, pr')$, the training objective of the QG model is to minimize the negative log-likelihood of the training data with respect to all parameters θ.

$$\begin{aligned}\mathcal{L} &= -\sum_{i=1}^{|D|} \log P(y^{(i)} \mid x^{(i)}; \theta) \\ &= -\sum_{i=1}^{|D|} \sum_{j=1}^{|y^{(i)}|} \log P(y_j^{(i)} \mid x^{(i)}, y_{<j}^{(i)}; \theta)\end{aligned} \tag{5}$$

Here, $\mathcal{L}$ denotes the loss function, which represents the model's performance on the training data; $y_j^{(i)}$ is the j-th word of the i-th sample; $y_{<j}^{(i)}$ represents all the words preceding the j-th word in the i-th sample; and θ stands for the model's parameters.

Component	Prompt
Document	{Article},{Answer},{Difficulty}
Role	You are an expert in the field of English studies.
Task Description	Easy-Level Questions: Recognition、Recall Medium-Level Questions: Understanding Hard-Level Questions: Application、Analysis、Creation、Evaluation
Task Requirements	The questions generated must meet the following conditions: Relevance: The generated questions must be directly relevant to the content of the provided article and answer, accurately reflecting its core ideas and key details. Clarity: All questions must be phrased in clear and concise language, following proper English grammar and language standards. Difficulty Levels: The questions must align with the specified difficulty levels (easy, medium, hard) as defined below. Question Design Guidelines: Questions must be well-formed and specific to the content of the article. Avoid ambiguous or overly complex wording. Ensure the question is logically consistent with the context provided by the article and the answer.

FIGURE 3. The prompt adopted in QG.

IV. EXPERIMENTAL SETTING

In this section, we conduct comprehensive experiments to demonstrate the effectiveness of our proposed method. Our experiments consist of two parts: construction of the difficulty-annotated dataset and Difficulty-Controllable QG. Section III-A annotates the difficulty of the EQG-RACE dataset, while the other parts verify the quality, difficulty accuracy, and diversity of the generated questions.

A. DATA ANNOTATION

For our experiments, we first trained a Diff-E model according to the LLM-AAL framework described in Section III-A to construct the difficulty-annotated dataset. We conducted our experiments on the EQG-RACE dataset [44], which holds extremely high educational and teaching value. The EQG-RACE dataset, derived from the RACE dataset, has removed data not suitable for the QG task. It comprises 18,501 training samples, 1,035 validation samples, and 950 testing samples, with each sample being a triplet of <article, answer, question>. Table 1 provides detailed information on the RACE and EQG-RACE datasets.

TABLE 1. Statistical information of the RACE and EQG-RACE datasets, where N and L represent the count and average length, respectively.

Dataset	Passage.N	Passage.L	Question.N	Question.L	Answer.L
RACE	27933	321.9	97687	10.00	5.30
EQG-RACE	12743	263.3	20486	10.87	6.36

To verify the superiority of maximum entropy sampling, we compared it with random sampling. As shown in Figure 4, the performance of the Diff-E model during the LLM-AAL iterative process using different sampling strategies is depicted. It is evident that the maximum entropy-based sampling strategy outperforms random sampling, with a faster convergence rate and higher Accuracy at the end of the iterations. Notably, the maximum entropy-based sampling strategy achieves better performance than random sampling with only half the amount of training data, making LLM-AAL more efficient and superior in performance. Additionally, we found that when the sample size reached 400, 600, and 800, the accuracy of the Diff-E model exceeded 0.85 in all cases, indicating that the Diff-E model can replace manual annotation.

We used the trained Diff-E model to annotate the difficulty levels of the EQG-RACE dataset. Figure 5 shows the distribution of samples after difficulty annotation. We named this dataset EQG-RACE-D and used it for the QG experiments. We found that the number of samples with medium difficulty is the largest. However, the distribution of easy: medium: hard samples in the training set, validation set, and test set are 4:6:1, 3.6:6:1, and 3.8:6:1, respectively. The samples are essentially in the same distribution, which is conducive to our subsequent QG experiments.

B. EVALUATION METRICS

We adopted a combination of automatic and human evaluations to assess the quality of the generated questions. For automatic evaluation, we used BLEU (1-4) [45] and ROUGE-L [46] to measure lexical matching and structural coherence. Additionally, BERT Score [26] and semantic Vector Similarity [47] were employed to provide a more comprehensive assessment from both shallow linguistic features and deep semantic aspects. The difficulty accuracy

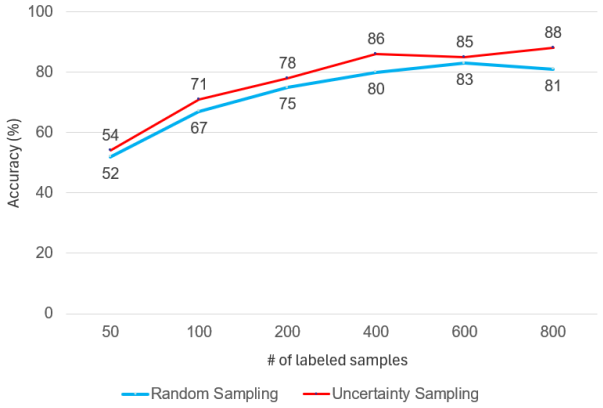


FIGURE 4. Accuracy of the Diff-E model using different sampling strategies during the LLM-AAL process (measured by accuracy).

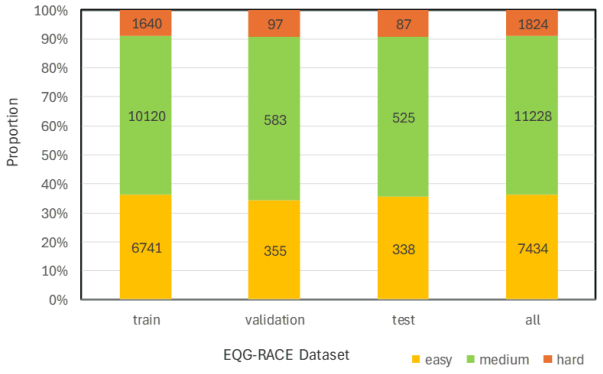


FIGURE 5. Statistics of question difficulty labels in the EQG-RACE dataset.

of the generated questions was evaluated using the Accuracy metric described in section III-A.

BLEU: Primarily used for machine translation and text generation tasks, measuring the n-gram overlap between generated text and reference text.

ROUGE-L: Designed for summarization and text generation tasks, evaluating the longest common subsequence match between generated text and reference text.

BERT Score: Based on contextual embeddings from PLMs like BERT, it calculates semantic similarity through fine-grained word alignment, capturing deep semantic and syntactic structures.

Vector Similarity: Computes the cosine similarity of overall text vectors using static word vectors (e.g., GloVe), measuring shallow lexical distribution matching.

C. METHODS COMPARED

We selected recent QG methods on the EQG-RACE dataset as baseline models for comparison with our proposed approach.

End-to-End Methods

Seq2Seq [48]: A classic sequence generation model based on encoder-decoder architecture, which dynamically computes the context vector through a global attention mechanism to mitigate the issue of information loss in long sequences.

Pointer-Generator [49]: A hybrid architecture that integrates a pointer network with LSTM, supporting the direct copying of vocabulary from the input sequence to the output sequence while combining a generation mechanism. This significantly enhances the coverage of rare words and proper nouns in tasks such as text summarization.

HRED [10]: A hierarchical recursive encoder-decoder model that captures long-term dependencies in multi-turn dialogues through session-level context encoding (hierarchical attention), commonly used in dialogue generation tasks.

Graph Neural Network Methods

AG-GCN [44]: The encoder uses a Graph Convolutional Network (GCN) to model structured inputs, while the decoder combines gated attention with a Maxout pointer network, suitable for generation tasks involving structured knowledge reasoning.

KPQDG [52]: This model employs a graph attention encoder and a dual gated attention layer to enhance encoding and reasoning capabilities based on the original passage.

Pre-trained Methods

Transformer [50]: A Seq2Seq model based on the self-attention mechanism, which abandons the recurrent structure and efficiently models long-range dependencies through parallel computation and positional encoding. It has become benchmark architecture for tasks such as machine translation and text generation.

ELMo-QG [51]: A question answering generation model that integrates ELMo dynamic word vectors with a Maxout pointer network. It enhances input encoding ELMo's context-sensitive representations and improves the accuracy of generated answers through a pointer mechanism that copies input fragments.

CEQG-T5 [47]: A unified text generation framework based on T5 (Text-to-Text Transfer Transformer), which formalizes the question answering generation task as a "text-to-text" transformation. It achieves end-to-end generation through a pretraining-finetuning paradigm and supports various input and output formats.

CEQG-BART [47]: A generation model based on BART (Bidirectional and Auto-Regressive Transformer), leveraging the combined advantages of a bidirectional encoder (similar to BERT) and an autoregressive decoder (similar to GPT). It performs excellently in tasks such as text summarization and question answering generation.

IEQG-FlanT5 [47]: An information extraction-based question answering generation model based on Flan-T5 (a T5 variant optimized for instruction tuning). It guides generation through task-specific prompts, focusing on extracting key information from structured data or text and transforming it into natural language questions.

D. IMPLEMENTATION DETAILS

1) LLM-AAL EXPERIMENTAL SETUP

Our LLM-AAL experiments were conducted on an NVIDIA RTX3060 (with 12GB of memory) using the PyTorch 2.5.0 and Transformers 4.46.1 frameworks for model training, with the hyperparameters used in the process as shown in Table 2.

TABLE 2. Statistics of the hyperparameters used in the LLM-AAL experiments.

Hyperparameter names	Type/Value
Learning rate	5e-5
Batch size	1
Epoch	4
Optimizer	AdamW
Learning rate scheduler	Cosine decay
Gradient accumulation steps	8
Temperature	0.7
Top_p	0.9

2) QG EXPERIMENTAL SETUP

Our QG experiments were conducted under the same experimental environment as LLM-AAL, with the hyperparameters used in the process as shown in Table 3.

TABLE 3. Statistics of the hyperparameters used in the QG experiments.

Hyperparameter names	Type/Value
Learning rate	5e-5
Batch size	2
Epoch	8
Optimizer	AdamW
Learning rate scheduler	Cosine decay
Gradient accumulation steps	8
Temperature	0.6
Top_p	0.9

V. RESULTS AND ANALYSIS

A. AUTOMATIC EVALUATION RESULTS

1) QUALITY EVALUATION

From Table 4, we can observe that our proposed model significantly outperforms traditional baselines across all metrics, demonstrating that the adapted LLM can effectively improve the quality of generated questions.

The data in Table 4 indicates that:

(a) Among the End-to-End Methods, the HRED model performs better than the baseline Seq2Seq and Pointer-Generator models, indicating that the global attention mechanism plays a critical role in the QG task.

(b) The AG-GCN model, based on Graph Neural Networks, significantly outperforms the end-to-end HRED model, demonstrating that GCN can effectively model structured input, and the gated attention mechanism and Maxout pointer network at the decoding end can effectively generate structured knowledge.

(c) The KPQDG model based on graph neural networks significantly outperforms the AG-GCN model, indicating that the graph attention mechanism can capture complex relationships between nodes and effectively enhance the representation capability of structured knowledge.

(d) The ROUGE-L score of the pre-trained CEQG-T5 model is superior to that of the KPQDG model, indicating that the questions generated by CEQG-T5 exhibit better semantic coherence. However, the BLEU-3/4 scores are slightly lower, suggesting that the questions generated by KPQDG place greater emphasis on the accuracy of key phrases and term sequences.

(e) At the model iteration level, the CEQG-BART model, based on BART, outper-forms CEQG-T5 in syntactic structure modeling due to its denoising autoencoder pre-training mechanism, while the IEQG-FlanT5 model, which incorporates instruction fine-tuning, further surpasses CEQG-BART by enhancing contextual semantic associations.

(f) At the model scaling level, the DL-CQG(Llama)$_{\text{LoRa}}$ framework, based on Llama-3.2-3B-Instruct, achieves optimal performance across all metrics, with its long-range dependency modeling capability and local language pattern capture ability significantly improving the semantic coherence and information completeness of the generated questions.

As shown in Table 5, the experimental results of our proposed framework under different LLMs and fine-tuning methods are presented. The results indicate that fine-tuning Qwen2.5-3B-Instruct using Freeze/LoRa outperforms fine-tuning BLOOM with Freeze/LoRa across all metrics, demonstrating that the questions generated by fine-tuning Qwen2.5-3B-Instruct exhibit superior grammatical precision and semantic accuracy. Similarly, fine-tuning Llama-3.2-3B-Instruct using Freeze/LoRa outperforms fine-tuning Qwen2.5-3B-Instruct with Freeze/LoRa across all metrics, suggest-ing that Llama-3.2-3B-Instruct's long-range information retention capability and precise understanding of linguistic details effectively enhance the quality of the generated questions. Notably, under the same LLM, the quality of questions generated via LoRa fine-tuning is generally better than that of Freeze fine-tuning. This indicates that full fine-tuning (Freeze) updates all parameters of the model, which, while ena-bling the model to learn the new task (e.g., question generation) effectively, tends to overwrite the general linguistic knowledge and reasoning abilities acquired during pre-training. In contrast, LoRa fine-tuning updates weights by introducing low-rank adapters while freezing most of the original model's parameters, meaning the core general knowledge base of the model remains intact. This approach results in higher-quality generated questions.

2) DIFFICULTY ACCURACY EVALUATION

We employed Accuracy to assess the Difficulty-Controllable of DL-CQG(Llama)$_{\text{LoRa}}$. Specifically, we used the Diff-E model to evaluate the difficulty of questions generated by DL-CQG(Llama)$_{\text{LoRa}}$ and then compared the evaluation results (difficulty level labels) with the true labels for consistency. The experimental results are shown in Figure 6.

Across the entire test set, the difficulty accuracy of questions generated by DL-CQG(Llama)$_{\text{LoRa}}$ reached 0.824, indicating that the DL-CQG(Llama)$_{\text{LoRa}}$ framework can effectively generate questions with Difficulty-Controllable. Further analysis revealed that the model performed best when generating ''easy'' questions, with an Accuracy of 0.858, suggesting that the model is more adept at handling factual tasks, such as information retrieval, simple question answering, or direct knowledge recall. The accuracy for generating ''medium'' difficulty questions was 0.819, which was stable but slightly lower than the ''easy'' questions, indicating that the model needed to provide some interpretation or reorganization of the information. The performance in generating ''hard'' questions was relatively poor, with an Accuracy of only 0.663, likely due to insufficient training data for the hard level, leading to the model failing to fully capture the characteristics of higher-order cognitive questions (such as analysis, evaluation, or creation).

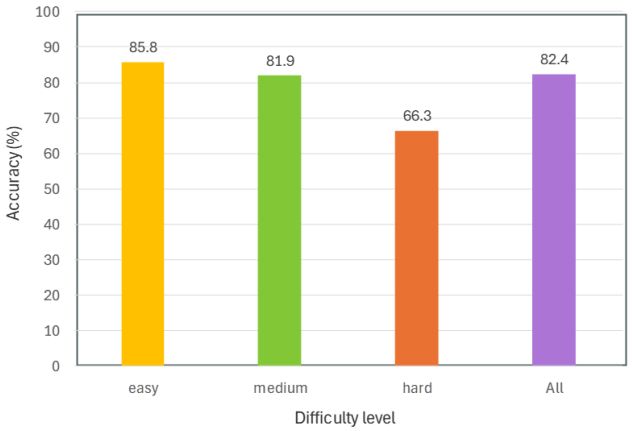


FIGURE 6. Validation results of the difficulty-controllable of QG by our model.

3) DIVERSITY EVALUATION

To evaluate the diversity of questions generated by the DL-CQG(Llama)$_{\text{LoRa}}$ framework, we adopt the commonly used text diversity metrics Distinct-1 and Distinct-2, which measure the richness of unigrams and bigrams, respectively, in the generated outputs. These metrics enable a more granular analysis of diversity across multiple questions produced by the model. We assess the diversity of the question set generated by the DL-CQG(Llama)$_{\text{LoRa}}$ framework using Distinct-1 and Distinct-2 scores. As shown in Figure 7, our framework achieves outstanding performance on both metrics, with average scores of 0.967 and 0.997, respectively. This result demonstrates that the DL-CQG(Llama)$_{\text{LoRa}}$ framework is highly effective at generating diverse questions. The performance advantage stems primarily from the synergy between the robust foundational capabilities of the Llama-3.2-3B-Instruct model and the LoRa fine-tuning strategy: Llama-3.2-3B-Instruct, as a large language model pre-trained on massive corpora, inherently possesses an extensive vocabulary and substantial linguistic expressive potential, while LoRa fine-tuning preserves the original model's inherent generative diversity to the greatest extent possible while adapting it to the downstream task.

TABLE 4. Comparative experimental results of the our framework. Here, the symbol B denotes BLEU, R denotes ROUGE, B-S denotes BERT Score, and V-S denotes Vector Similarity.

Models	B-1	B-2	B-3	B-4	R-L	B-S	V-S
Seq2Seq	23.46	10.24	6.39	4.75	23.82	N/A	N/A
Pointer-Generator	28.96	14.33	8.82	5.99	30.02	N/A	N/A
HRED	28.69	15.14	9.42	6.16	32.70	N/A	N/A
Transformer	28.93	15.20	9.45	6.25	32.43	N/A	N/A
ELMo-QG	33.95	18.55	11.93	8.23	33.26	N/A	N/A
AG-GCN	35.10	21.08	15.19	11.96	34.24	N/A	N/A
KPQDG	41.97	32.39	26.65	22.58	42.39	N/A	N/A
CEQG-T5	N/A	N/A	25.19	20.60	46.48	76.96	88.95
CEQG-BART	N/A	N/A	26.59	21.99	47.95	77.55	89.08
IEQG-FlanT5	N/A	N/A	27.88	22.90	49.70	78.28	90.04
DL-CQG(Llama)$_{\text{LoRa}}$	**66.16**	**56.31**	**49.62**	**45.20**	**50.01**	**81.26**	**97.04**

TABLE 5. A Comparative Experimental Study of Different LLMs and Fine-tuning Methods in the our framework.

Models	B-1	B-2	B-3	B-4	R-L	B-S	V-S
DL-CQG(BLOOM)$_{\text{Freeze}}$	60.62	48.27	39.02	33.41	38.05	69.82	84.94
DL-CQG(BLOOM)$_{\text{LoRa}}$	63.27	53.25	45.64	40.75	45.14	74.12	92.36
DL-CQG(Qwen)$_{\text{Freeze}}$	63.55	52.30	43.84	38.62	42.08	72.18	90.20
DL-CQG(Qwen)$_{\text{LoRa}}$	65.25	55.41	48.08	43.34	47.06	76.27	95.42
DL-CQG(Llama)$_{\text{Freeze}}$	63.96	51.97	43.91	38.84	43.51	75.36	90.88
DL-CQG(Llama)$_{\text{LoRa}}$	**66.16**	**56.31**	**49.62**	**45.20**	**50.01**	**81.26**	**97.04**

B. ABLATION EXPERIMENTS

To verify the effectiveness of our prompt and LoRa fine-tuning strategies, we conducted ablation experiments. The results of the ablation experiments are presented in table 6. As shown in table 6, removing either the prompt or the LoRa fine-tuning leads to a decline in the performance of the DL-CQG(Llama)$_{\text{LoRa}}$ framework. Specifically, removing the prompt has a relatively minor impact on the overall performance of the model. Although the model can still generate reasonable questions, it weakens the model's ability to precisely control the output structure. In contrast, removing LoRa fine-tuning significantly impairs the model's capability, resulting in questions with insufficient relevance that fail to meet the demands of high-quality QG.

C. HUMAN EVALUATION RESULTS

To more comprehensively evaluate the quality of the generated questions, we further validated them through manual assessment. Specifically, we compared the questions generated by the DL-CQG(Llama)$_{\text{LoRa}}$ framework with those produced by the ELMo-QG and AG-GCN models. We randomly sampled 100 instances each and invited three educational assessment experts to evaluate them. The scoring range was from 0 to 10, with higher scores indicating better quality of the generated questions. The evaluation criteria were as follows:

Fluency: Whether the question is grammatically correct and fluent.

Relevancy: Whether the question is semantically relevant to the passage.

Answerability: Whether the question can be answered with a correct answer.

Difficulty: The subjective difficulty assessment of the question, where lower scores indicate lower difficulty.

To ensure the reliability and validity of the scoring, we used the intraclass correlation coefficient (ICC) (applicable to continuous variable tasks where multiple evaluators independently score the same set of objects) to measure the consistency of the three experts' ratings. We calculated the average scores of the three evaluators and the ICC values, with the results shown in table 7. The ICC values for DL-CQG(Llama)$_{\text{LoRa}}$, ELMo-QG, and AG-GCN exceeded 0.75 (ICC values between 0.75 and 0.9 indicate good consistency), demonstrating that the experts' ratings had good agreement and the results were reliable. We found that the questions generated by DL-CQG(Llama)$_{\text{LoRa}}$ scored the highest in fluency, relevance, and answerability, while AG-GCN showed comparable competitiveness. ELMo-QG performed poorly, which may be due to its limited contextual modeling capability.

To verify the accuracy of the Diff-E model, we selected 50 questions of difficulty levels "easy", "medium", and "hard" respectively from the questions generated by the DL-CQG(Llama)$_{\text{LoRa}}$ framework. These questions were then evaluated by domain experts based on Bloom's Taxonomy. The evaluation results, as shown in Figure 8, indicate that the accuracy of the DL-CQG(Llama)$_{\text{LoRa}}$ framework in generating questions labeled "easy", "medium", and "hard" is 0.86, 0.82, and 0.70, respectively, demonstrating the effectiveness of the Diff-E model.

Additionally, we employed the same method to validate the difficulty accuracy of the questions generated by

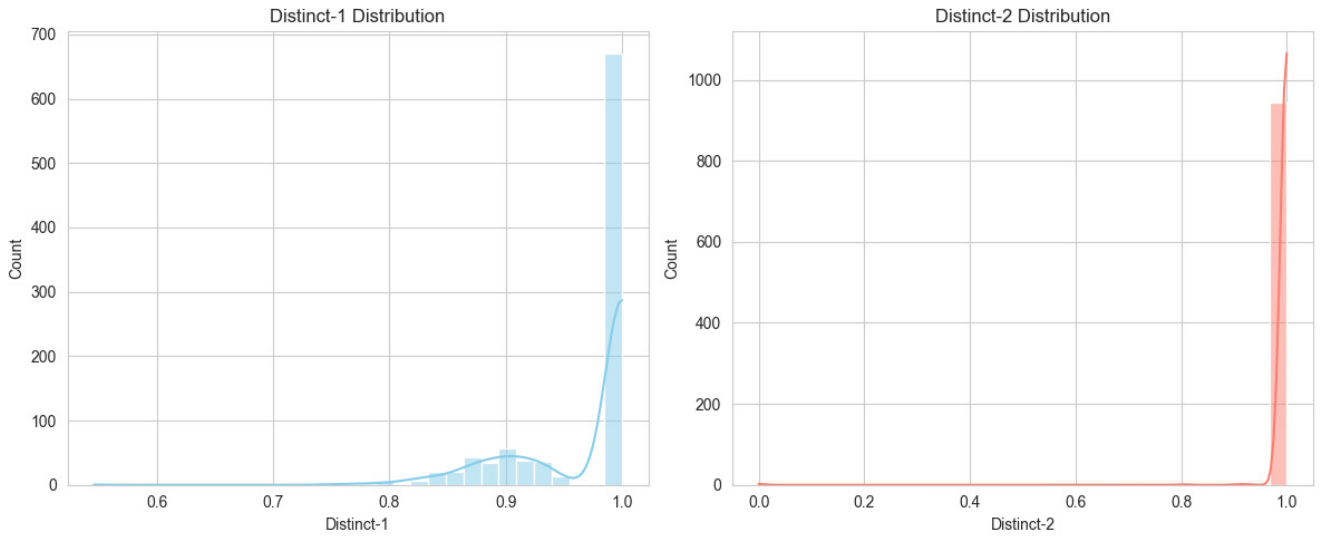


FIGURE 7. Diversity evaluation of QG by our framework.

TABLE 6. Ablation experiment results of QG by the our framework. Here, the symbol B denotes BLEU, R denotes ROUGE, B-S denotes BERT Score, and V-S denotes vector similarity.

Models	B-1	B-2	B-3	B-4	R-L	B-S	V-S	Accuracy
DL-CQG(Llama)$_{\text{LoRa}}$	**66.16**	**56.31**	**49.62**	**45.20**	**50.01**	**81.26**	**97.04**	**0.824**
Without prompt	65.43	55.48	48.68	44.17	49.52	79.02	92.30	0.796
Without LoRa fine-tuning	13.39	10.38	7.64	5.85	5.11	43.12	67.73	0.143

TABLE 7. Evaluation results of manually assessing QG by different models (covering three metrics: Fluency, Relevancy, and Answerability). The ICC values range from 0 to 1.

Models	Fluency	Relevancy	Answerability	ICC
ELMo-QG	7.21	6.71	7.34	0.81
AG-GCN	7.94	7.05	8.18	0.80
DL-CQG(Llama)$_{\text{LoRa}}$	**8.63**	**7.96**	**8.65**	0.86

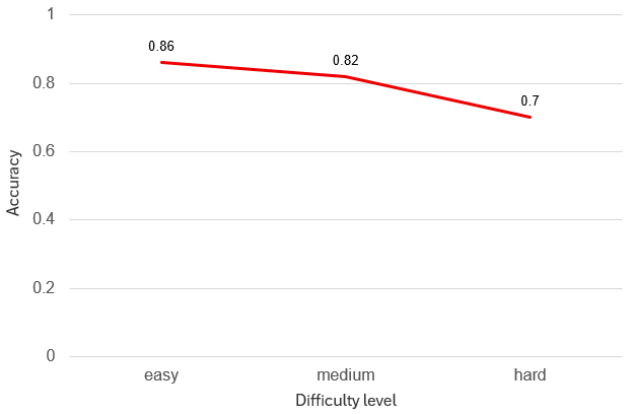


FIGURE 8. Human evaluation the accuracy of the Diff-E model's difficulty level on evaluation questions.

DL-CQG(Llama)$_{\text{LoRa}}$. We calculated the average difficulty scores assigned by the three evaluation experts to the sampled questions. Figure 9 illustrates the relationship between the specified difficulty levels of the questions and the average difficulty scores given by the three experts. It is evident that DL-CQG(Llama)$_{\text{LoRa}}$ can generate questions with the specified difficulty levels. Notably, the average difficulty scores assigned by the three experts to the generated questions were largely consistent, indicating the reliability of the evaluation results.

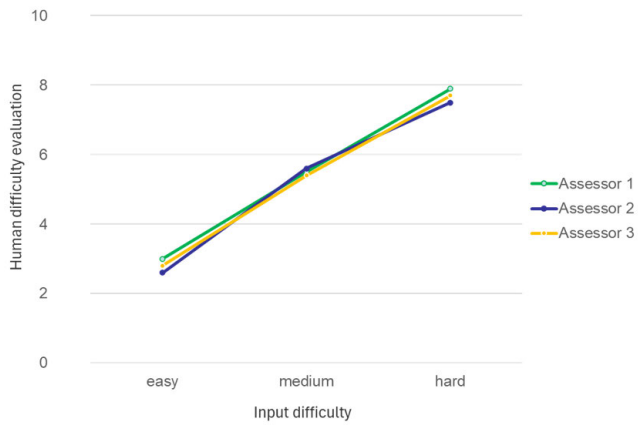


FIGURE 9. The relationship between the specified difficulty levels of the questions and the average difficulty scores assigned by the three evaluation experts.

TABLE 8. We use different colors and underlines to highlight the relevant information between the QG by different models and the article.

Article: Alice 's mother died when she was five. although her brothers and sisters were loving and caring, their love couldn't take the place of a mother 's. in 1925 alice became my mother and told me that her family couldn't even afford her a doll. one afternoon in December 1982, when i was getting ready for christmas, I suddenly decided to buy two dolls, one for my five-year-old daughter, katie, and one for my old mother. things went smoothly when a friend told me that his dad, who played santa claus in my area, would be willing to make a visit on christmas morning to our home with the gifts! knowing that my parents would also come to my house, i began to get ready for the most memorable day of my mother 's life. Christmas day arrived and so did santa claus at the planned time.... She couldn't speak but only held tightly in her arms the doll she had waited fifty-seven years to receive as tears of joy flew down her face. That doll made my mother the happiest child.

Answer: because her family was very poor.

Question: Why was Alice unable to receive a doll during her childhood in 1925? (Difficulty: medium)

ELMo-QG: Why didn't Alice get a doll as a child?

AG-GCN: What prevented Alice's family from giving her a doll in her childhood?

DL-CQG(Llama)$_{\text{LoRa}}$: How does the article explain Alice's 57-year wait to receive her first doll?

TABLE 9. We used blue to highlight the information in the DL-CQG(Llama)$_{\text{LoRa}}$ generated questions related to the article content and underlined to mark the information in the truth questions related to the article content.

Article: The study considered three general styles of parenting. Authoritative parents are demanding and controlling while also being warm and sensitive to their children's needs. Authoritarian parents, on the contrary, are demanding and controlling without the appearance of caring, attachment, and receptiveness. They take a 'my way or the highway' approach to their kids. Permissive parents, the third group, have warm and receptive qualities, but they set few boundaries and carry out few rules. Using data on nearly 600 kids from an ongoing study of middle school and high school students in New Hampshire, researchers from the University of New Hampshire were able to link 'my way or the highway' parenting with more violent behaviors like robbery, drug-taking, and attacking someone else with the intention of hurting or killing. Firm but loving parenting, on the other hand, led to fewer lawbreakers. Permissive parenting, surprisingly, didn't seem to make much of a difference either.

To explain the link between parenting style and behavior in kids, the researchers suggested that what matters most is how reasonable kids think their parents' power is. This sense comes when kids trust that their parents are making the best decisions for them and believe that they need to do what their parents say even if they don't always like how their parents are treating them. When kids respect the power of their parents, the researchers reported in the Journal of Adolescence, their behavior is better. Previous research has also linked firm but caring parenting with kids who have more self-control and independence.

Answer: They trust parents make good decisions for them.

Question: Why do kids obey authoritative parents? (Difficulty: hard)

DL-CQG(Llama)$_{\text{LoRa}}$: What psychological mechanism underlies children's compliance with authoritative parenting styles?

D. CASE STUDY

(1) Table 8 presents a performance comparison analysis of the DL-CQG(Llama)$_{\text{LoRa}}$ framework and baseline models (ELMo-QG, AG-GCN) on the QG task based on the article "The Long-Awaited Doll: Alice's 57-Year Christmas Miracle". The comparison reveals the following findings: First, the ELMo-QG model exhibits noticeable text fragment copying in its generated questions, such as directly quoting the phrase "couldn't afford her a doll" from the original text. This generation approach only requires simple literal matching to answer, reflecting the model's limitations in semantic understanding. This phenomenon may stem from its pointer mechanism's over-reliance on text fragment copying and its lack of ability to model long-range semantic relationships. In contrast, the AG-GCN model outperforms ELMo-QG. Its performance improvement arises from two aspects: the graph attention mechanism effectively captures key entity relationships, and the decoder's gating mechanism enhances the fluency of the generated questions. However, the model's graph attention mechanism focuses more on the topological structure between entities (e.g., family → poverty → doll) during relation extraction but shows insufficient sensitivity to non-entity information, which may lead to ambiguous questions. The DL-CQG(Llama)$_{\text{LoRa}}$ framework performs the best. This model demonstrates the following advantages: 1) the ability to deeply integrate key temporal clues (e.g., "57-year wait"); 2) causal reasoning capabilities rather than simple fact extraction; and 3) the ability to generate questions that align with specified difficulty labels. These advantages primarily benefit from the powerful contextual semantic understanding of its underlying Llama-3.2-3B-Instruct model. Notably, through prompt engineering and fine-tuning techniques, the model can explicitly learn the instruction patterns for medium difficulty labels, thereby maintaining consistency with the specified difficulty during decoding.

(2) Table 9 presents the limitations of the DL-CQG (Llama)$_{\text{LoRa}}$ framework in the QG task based on the article "Parenting Styles and Adolescent Behavior: Authoritarian Approaches Associated with Higher Violence,

Study Shows''. Through in-depth analysis, this study reveals the following findings: First, there is a bias in matching question difficulty. Although the questions generated by the model were correctly labeled as hard difficulty (as they involve the abstract concept of ''psychological mechanism''), the actual answer, ''They trust parents make good decisions for them,'' is merely a direct statement about a specific trust relationship and fails to meet the preset abstract cognitive level of the question. This mismatch in cognitive levels between the question and answer reflects the model's inadequacy in difficulty control. Second, in terms of semantic consistency, the model exhibits a tendency toward mechanical associations. The analysis indicates that the model may have superficially linked the hard difficulty label with specific abstract concepts without deeply understanding the required semantic depth of the question, resulting in flaws in the semantic coherence of the generated questions.

VI. CONCLUSION

In this work, we proposed a unified framework for Difficulty-Controllable QG, named DL-CQG. Our framework defines question difficulty levels based on Bloom's theory. We use LLM-AAL iterative loop to train a model that can replace humans to evaluate the difficulty of the question and realize the difficulty annotating of EQG-RACE dataset. On this basis, we adapted Difficulty-Controllable QG tasks by fine-tuning LLMs (Qwen, Llama) with prompts and LoRa. Experiments on the EQG-RACE dataset demonstrated the effectiveness of our proposed framework. Our DL-CQG(Llama)$_{\text{LoRa}}$ framework improved the evaluation metrics to 45.20 (BLEU-4), 50.01 (ROUGE-L), and 81.26 (BERT Score), respectively. This work provides a unified paradigm for achieving Difficulty-Controllable QG under low-cost conditions. In the LLM-AAL process, we manually annotated only 800 samples; in QG, we selected models with fewer parameters (BLOOM, Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct) and trained the models on RTX-3060 (with 12G of memory).

In future work, we aim to make improvements in two areas: (1) Enhancing the model's ability to generate higher-cognitive questions by improving sample quality. (2) Exploring the deep integration of knowledge graphs and LLMs, utilizing the rich conceptual hierarchical relationships in knowledge graphs (such as hierarchical relationships, causal associations, etc.) to construct a fine-grained cognitive difficulty evaluation system, thereby achieving hierarchical and fine-grained control of question difficulty.

APPENDIX. ANNOTATION GUIDELINES

Easy – Fact Extraction

The task only requires recognizing or recalling learned facts, terminology, symbols, and concepts.

Typical verbs: list, define, recall, match, select, label, restate.

Guiding question: ''Does the answer exist directly in memory or the provided material?''

Example: ''What are Newton's three laws of motion?''; ''Identify the mitochondria in the diagram.''

Medium – Meaning Construction

The task requires transforming, interpreting, or inferring information, but does not go beyond the boundaries of existing knowledge.

Typical verbs: summarize, exemplify, compare (at a basic level), classify, explain, paraphrase.

Guiding question: ''Do you need to reorganize the information in your own words?''

Example: ''Explain why Newton's first law is also called the law of inertia.''; ''Compare the similarities and differences between photosynthesis and respiration (only listing known distinctions).''

Hard – Higher-Order Operations

The task involves applying knowledge in new contexts, systematic analysis, critical evaluation, or original construction.

Typical verbs:

Apply: solve, calculate, demonstrate, operate

Analyze: distinguish, attribute, structure, relate, contrast (in-depth)

Evaluate: judge, verify, critique, recommend

Create: design, construct, hypothesize, invent, compose

Guiding question: ''Does the task require reasoning, decision-making, or producing something beyond directly taught methods or cases?''

Example: ''Use Newton's laws to calculate the acceleration of an object on an inclined plane.''; ''Analyze possible systematic errors in this experimental data.''; ''Evaluate the impact of different transportation policies on urban carbon emissions.''; ''Design a simple device to verify the conservation of energy.''

APPENDIX. REPRESENTATIVE EXAMPLES OF DIFFICULTY ANNOTATION

Easy: Where did Alice's mother live when she was a child?

Corresponding Article Context: The article mentions the mother's childhood place of residence.

Medium: Why did the author decide to buy two dolls instead of one?

Corresponding Article Context: The article needs to analyze the dual motivations of the author's actions (as a daughter and as a mother).

Hard: How might the story have changed if Santa had forgotten the second gift?

Corresponding Article Context: It is necessary to conduct counterfactual reasoning based on the emotional tone of the article.

DATA AVAILABILITY

The data supporting the findings of this study have been deposited in relevant repositories. The persistent access link for the EQG-RACE dataset is https://github.com/jemmryx/EQG-RACE.

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XIAOHUI DONG received the Ph.D. degree from Northwest Normal University, Gansu, China. He is currently an Associate Professor and a Master's Supervisor with the College of Computer Science and Engineering, Northwest Normal University. His research interests include artificial intelligence, knowledge graph, and intelligent education.



ZHENGLUO LI is currently pursuing the degree with the College of Computer Science and Engineering, Northwest Normal University. His research interests include artificial intelligence, natural language processing, and intelligent education.